

Isomerism

Configurational Isomers

Configurational isomers are stereoisomers that cannot be converted into one another by rotation around a single bond.
(geometric isomers and optical isomers).



geometric isomers
cis-trans isomers

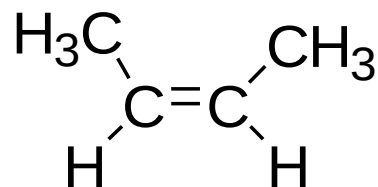
optical isomers
isomers that contain
chirality centers

geometric isomers

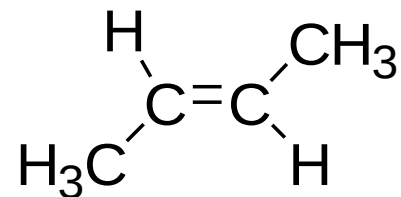
cis-trans isomers

2-butene : $\text{CH}_3\text{CH}=\text{CHCH}_3$

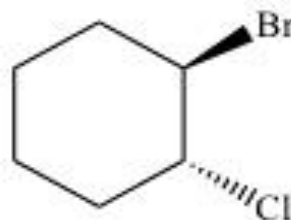
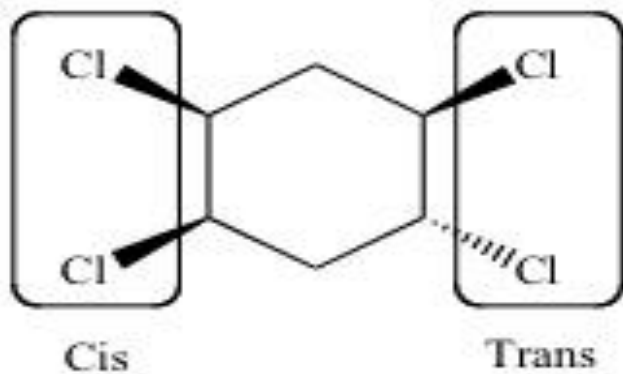
2 geometric isomers



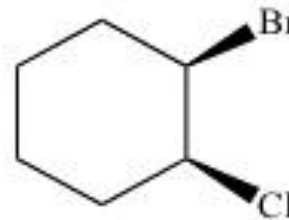
cis-2-butene



trans-2-butene

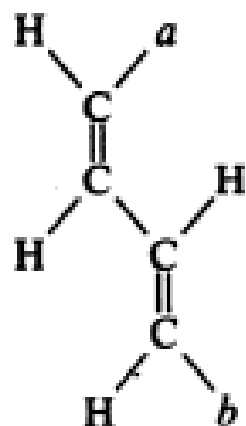


⊥

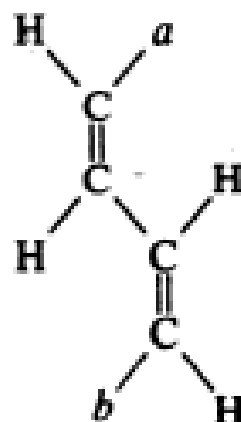


C

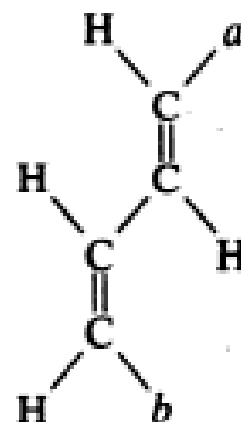
If a compound has two double bonds, four geometrical isomers are possible



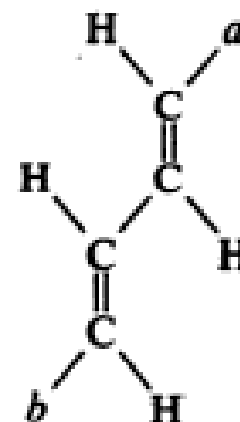
cis-trans



cis-cis



trans-cis



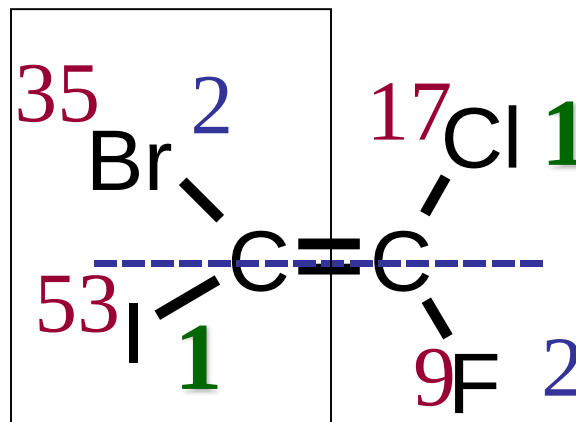
trans-trans

E and Z nomenclature

E and Z Based on Priority:

Cahn-Ingold-Prelog Rules

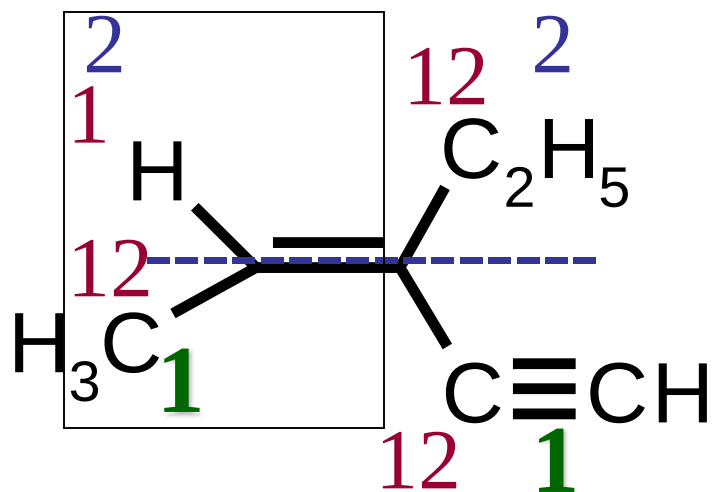
- Assign a priority number to each group attached to the chiral carbon.
- Atom with highest atomic number assigned the highest priority #1.
- In case of ties, look at the next atoms along the chain.
- Double and triple bonds are treated like bonds to duplicate atoms.



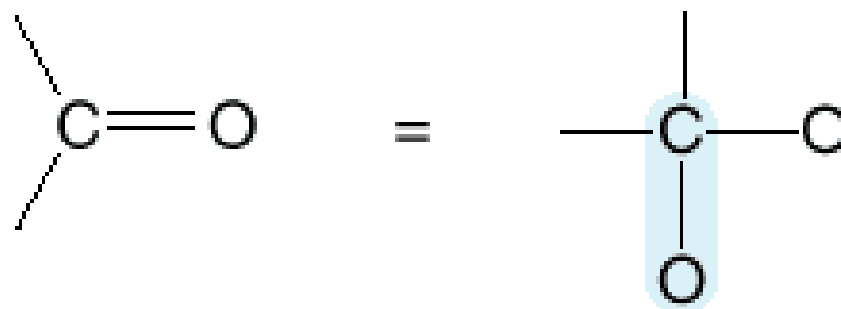
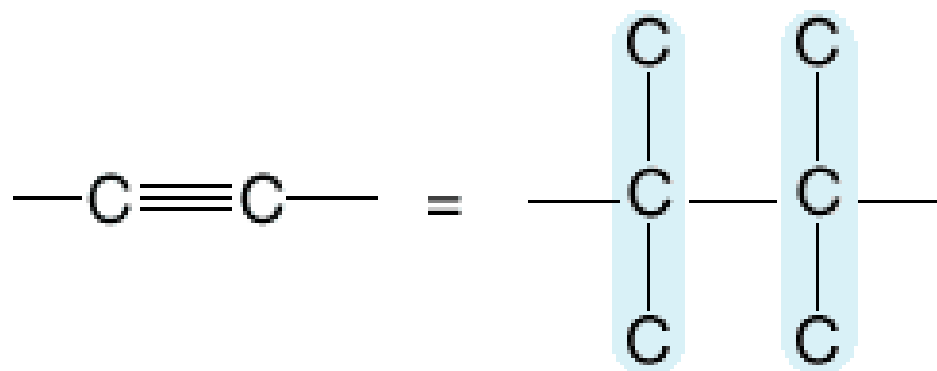
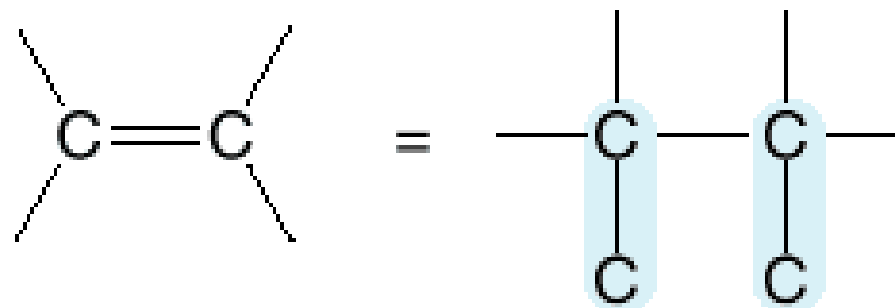
Higher priority at the **opposite** side of pi bond (**E**)

Higher priority at the **same** side of pi bond (**Z**)

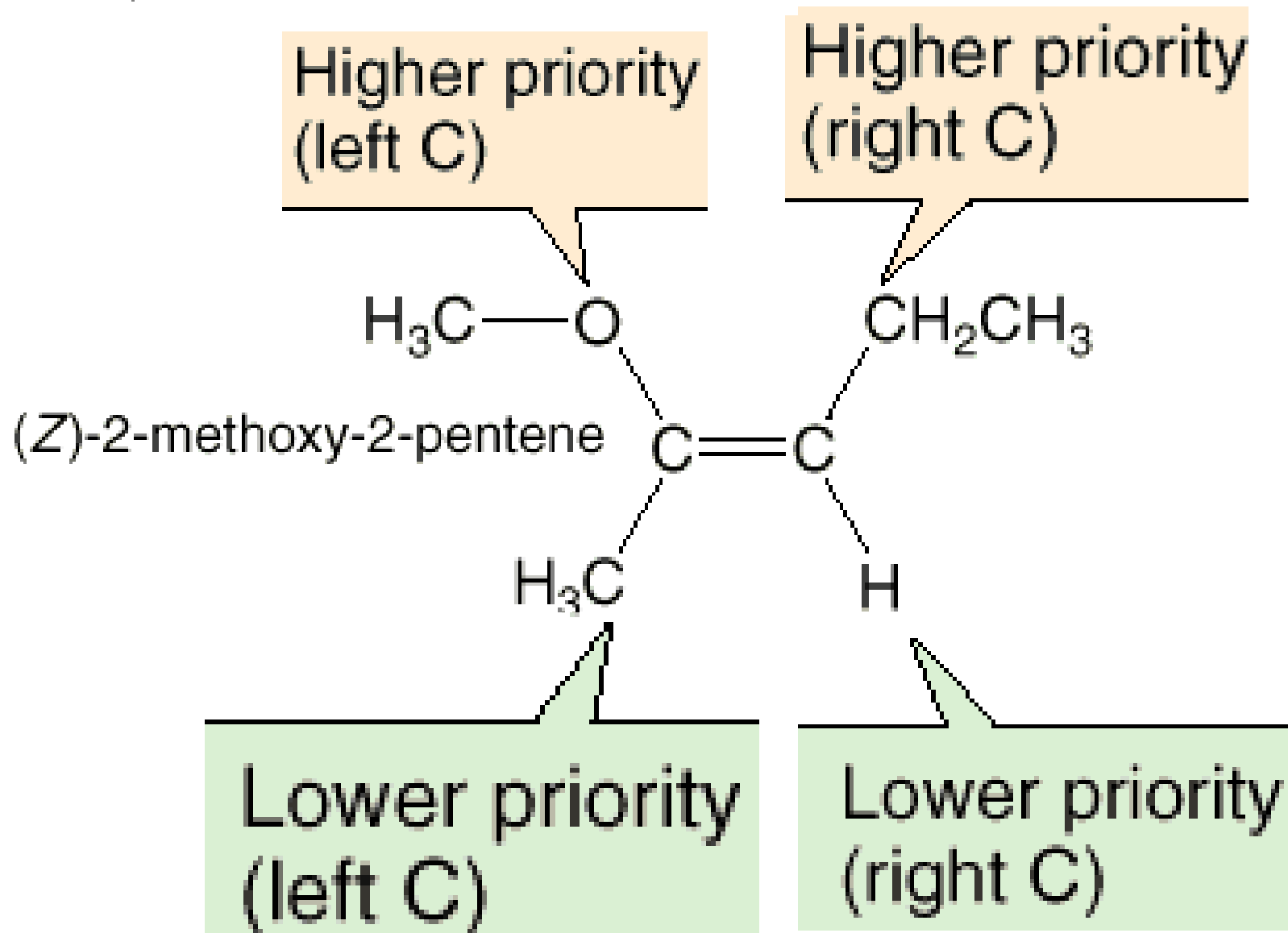
(E)-1-bromo-2-chloro - 2-fluoro-1-iodoethene



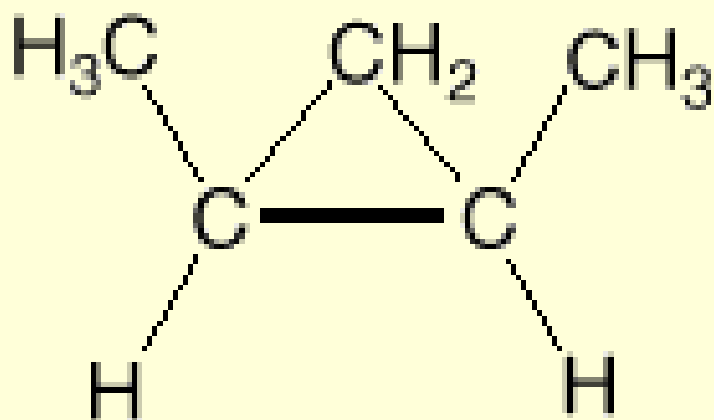
(Z)-3-ethyl-2-pentene-4-yne



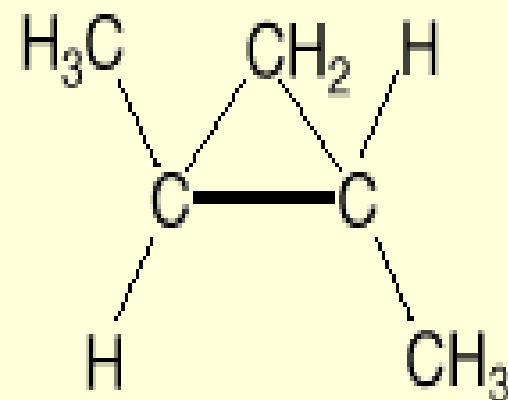
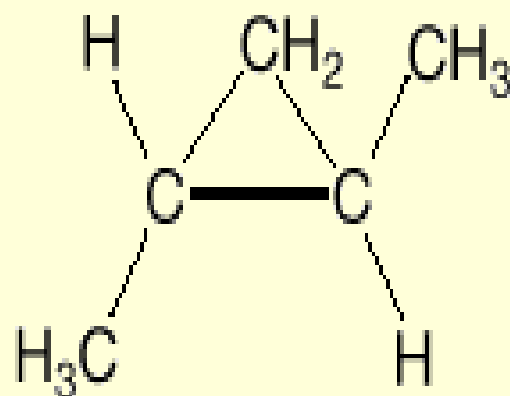
The two high priority groups are on the same side — this molecule is (*Z*)-



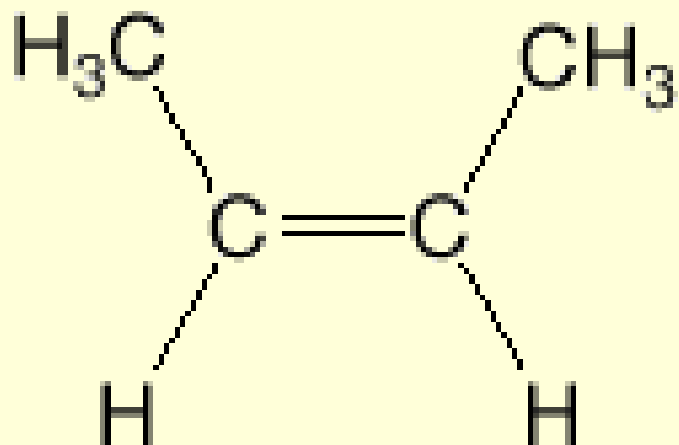
Geometric Isomers



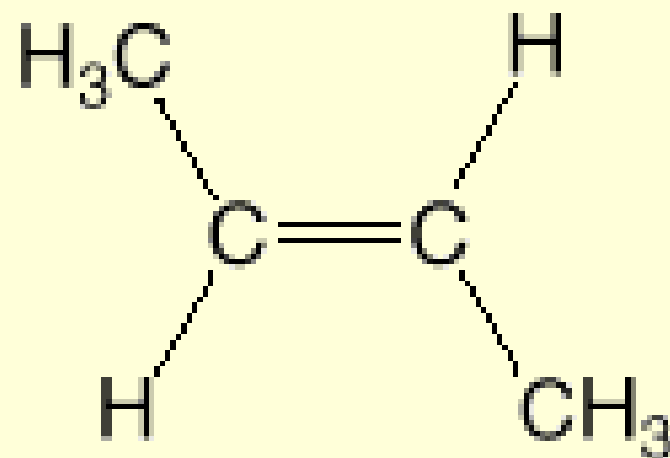
***cis*-1,2-Dimethylcyclopropane**



Two representations of
***trans*-1,2-dimethylcyclopropane**



***cis*-2-Butene [or (*Z*)-2-butene]**

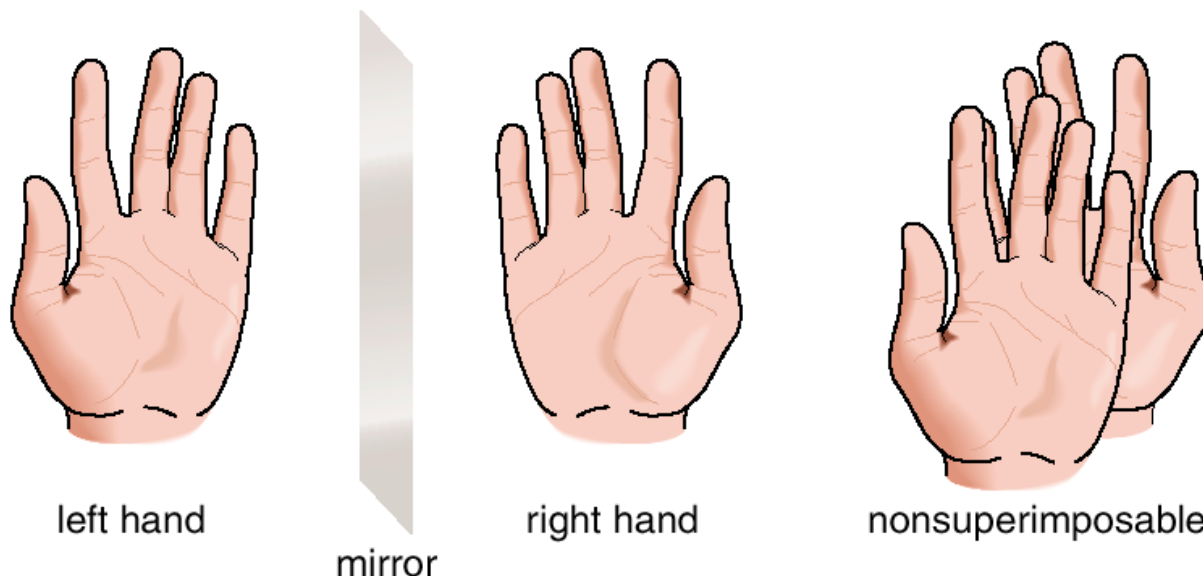


***trans*-2-Butene [or (*E*)-2-butene]**

Optical Isomerism

Some objects are not the same as their mirror images (technically, they have no plane of symmetry)

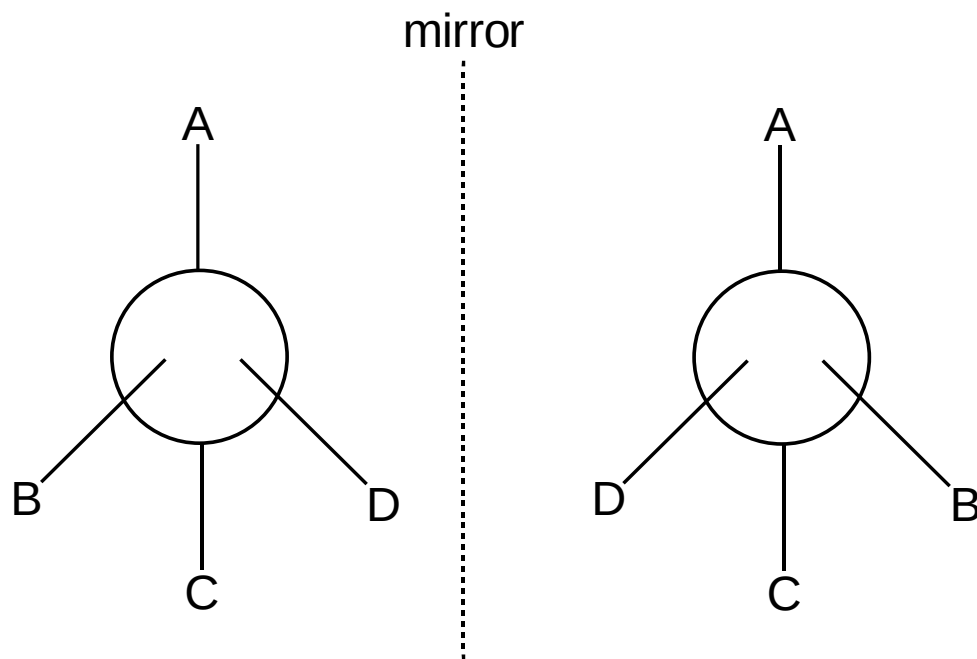
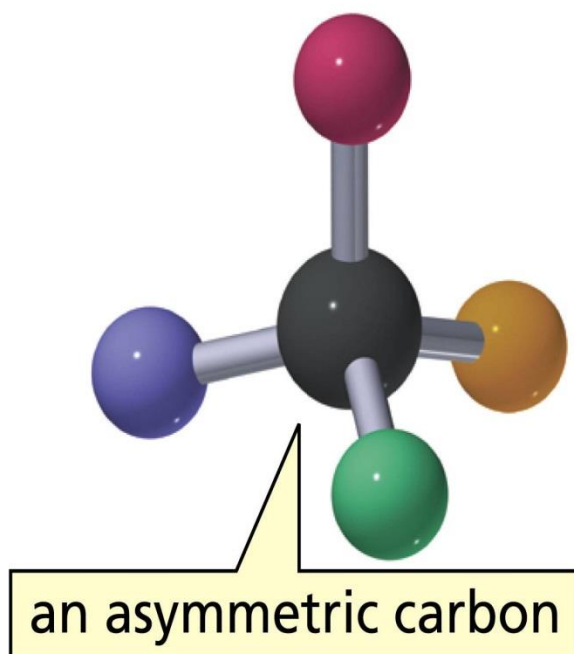
A right-hand glove is different than a left-hand glove. The property is commonly called “handedness”



- A molecule (or object) that is *not* superimposable on its mirror image is said to be *chiral*.

Chirality

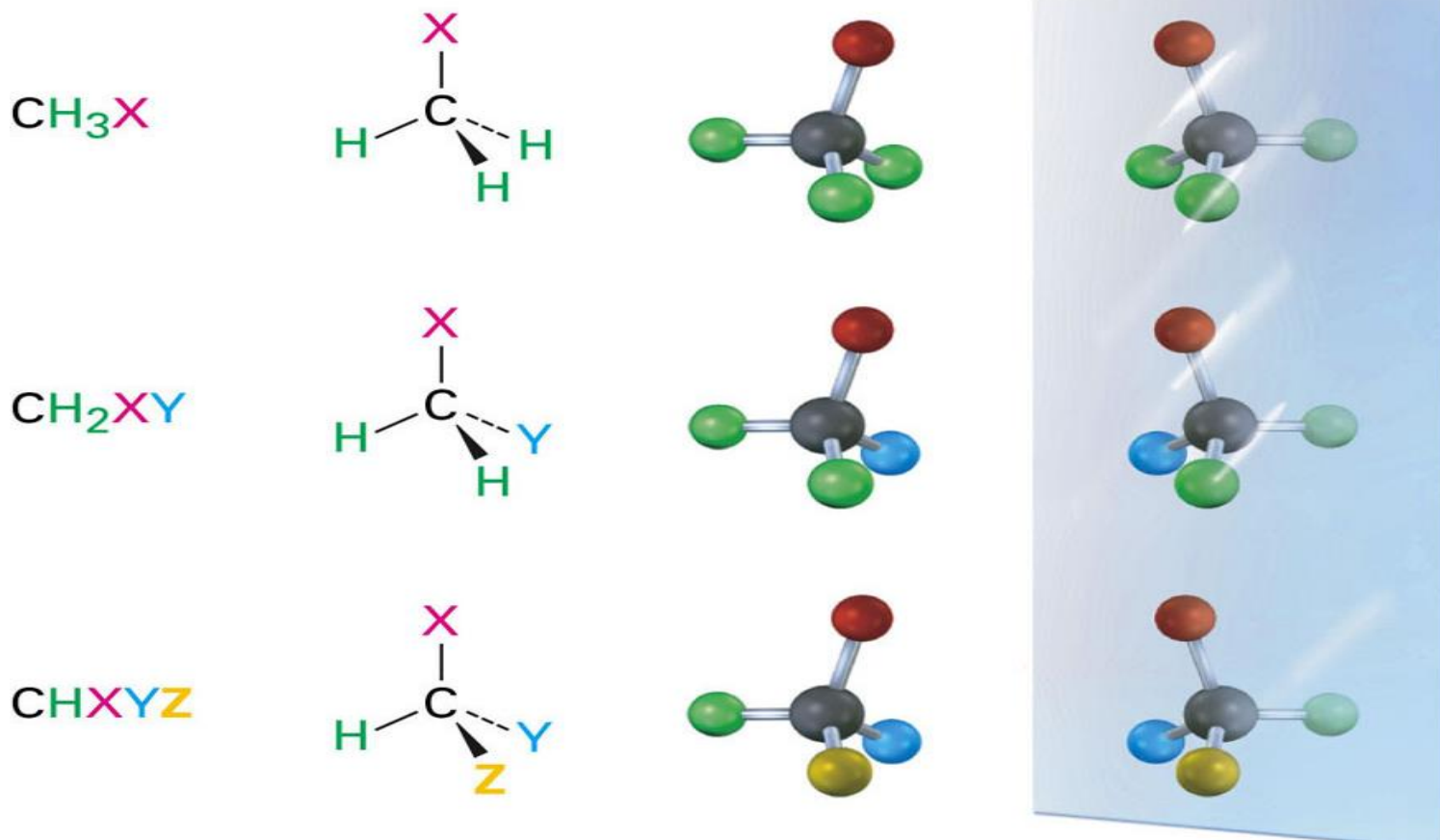
A chiral molecule is a type of molecule that has a non-superposable mirror image. The property is most often the cause of the presence of an asymmetric carbon atom

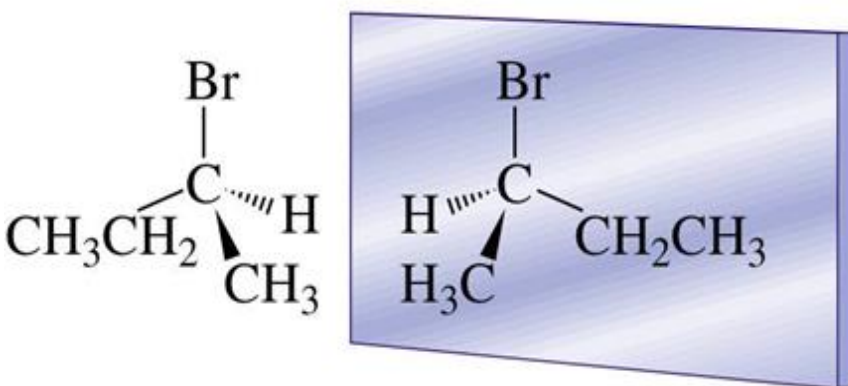


The two are non-superimposable, mirror images. Such isomers are called enantiomers.

Examples of Enantiomers

- Molecules that have one carbon with 4 different substituents have a non-superimposable mirror image
- **Enantiomers** = non-superimposable mirror image stereoisomers
- Enantiomers have identical physical properties (except for one)
- Build molecular models to see this

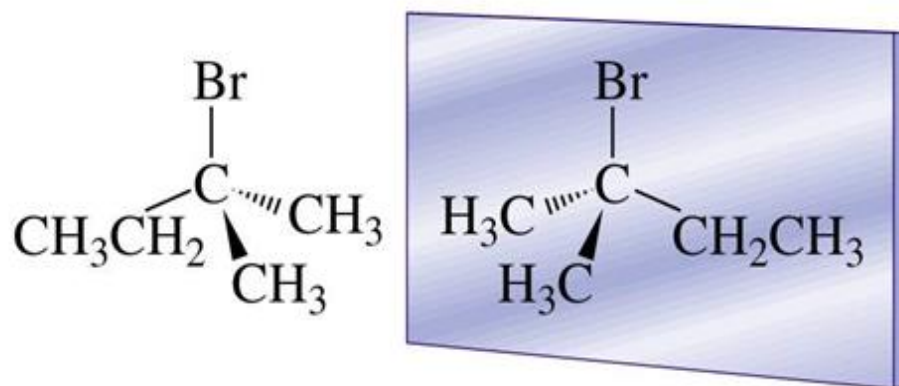




a chiral
molecule

nonsuperimposable
mirror image

enantiomers



an achiral
molecule

superimposable
mirror image

identical molecules

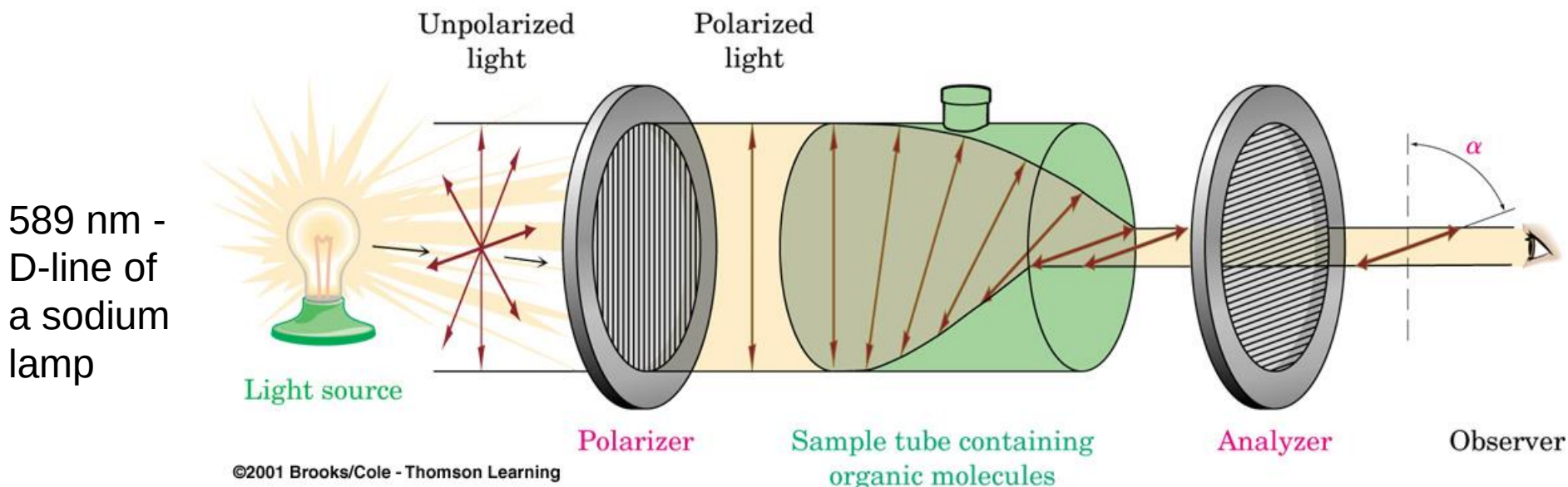
Optical Activity –

molecules enriched in an enantiomer will rotate plane polarized light are said to be *optically active*. The optical rotation is dependent upon the substance, the concentration, the path length through the sample, and the wavelength of light.

Rotation, in degrees, is $[\alpha]$

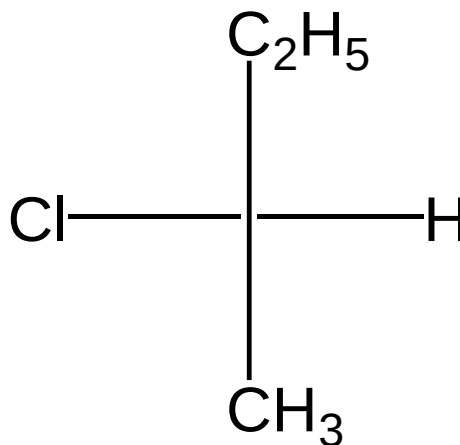
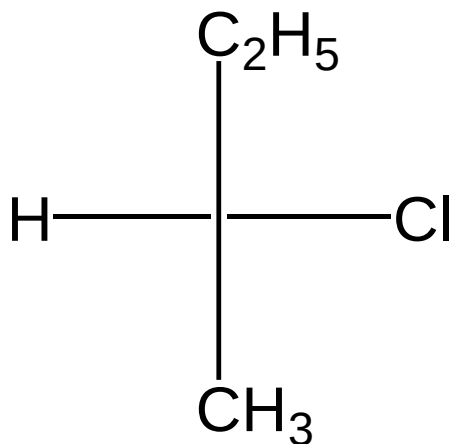
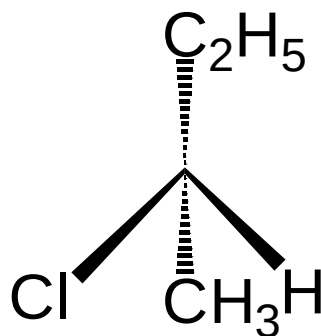
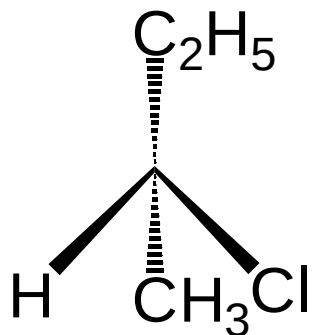
Clockwise (+) = dextrorotatory; Anti-clockwise (-) = levorotatory

Polarimeter



Plane polarized light: light that oscillates in only one plane

How do we “draw” a chirality centre?

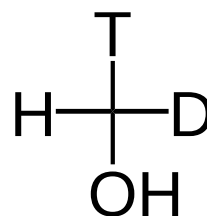
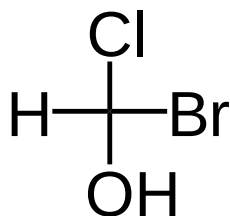


Fischer Projection

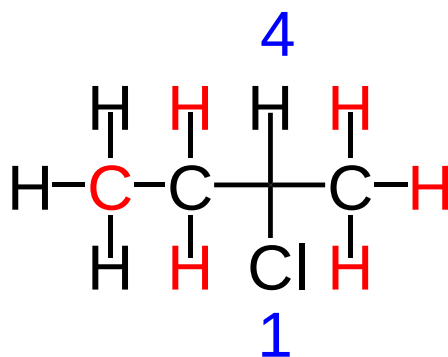
Cahn - Prelog - Ingold rules for nomenclature

Step 1: assign a priority to the 4 atoms or groups of atoms bonded to the tetrahedral stereogenic centre:

1. If the 4 atoms are all different, priority is determined by atomic number. The atom of higher atomic number has the higher priority.

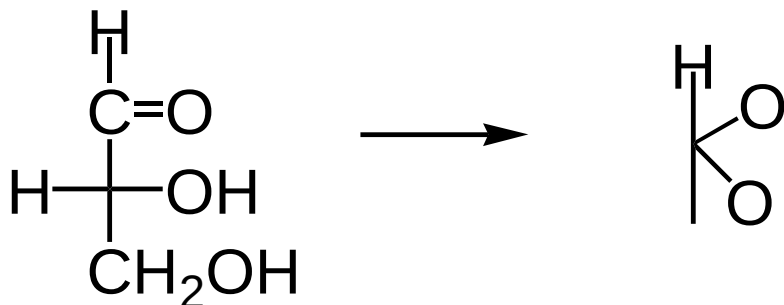


2. If priority cannot be determined by (1), it is determined by a similar comparison of atoms working out from the stereocentre.



In the methyl group, the second atoms are H, H, H whereas in the ethyl group, they are C, H, H. The priority sequence is therefore Cl, C₂H₅, CH₃, H.

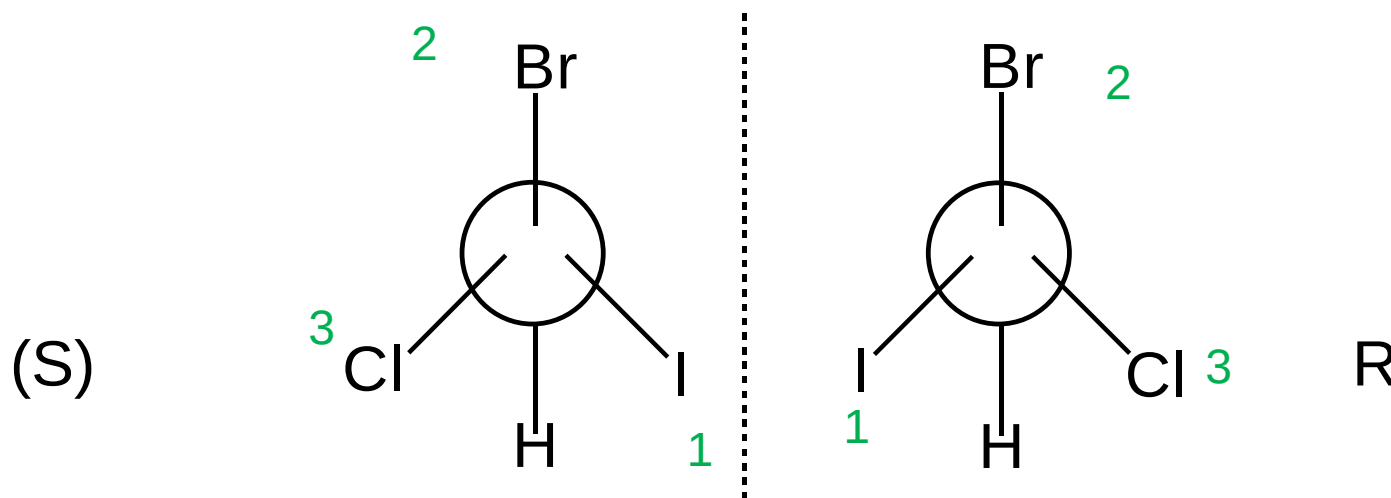
3. A double or triple bond to an atom, **A**, is considered as equivalent to two or three single bonds to **A**:



The sequence is therefore -OH, -CHO, -CH₂OH, -H.

Step 2

Arrange the molecule so that the group of lowest priority is pointing away from you and observe the arrangement of the remaining groups:

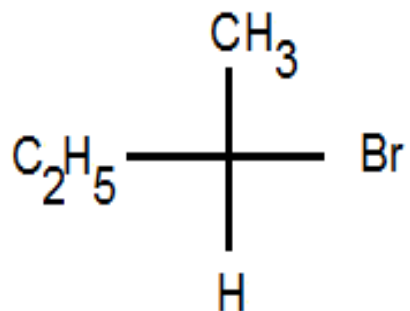


If, on going from the group of highest priority to that of second priority and then to the group of third priority, we go in a clockwise direction, the enantiomer is designated (R).

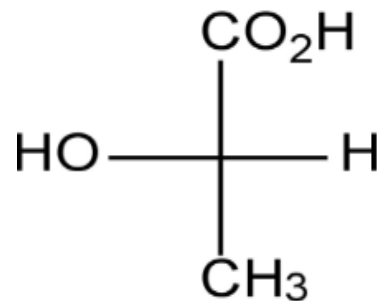
If the direction is counterclockwise, the enantiomer is designated (S).

Thus the complete name for one of the enantiomers of 2-chlorobutane is (R)-2-chlorobutane

R or S?

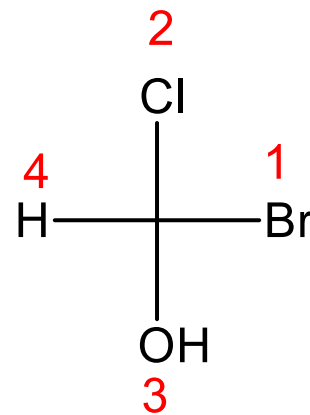
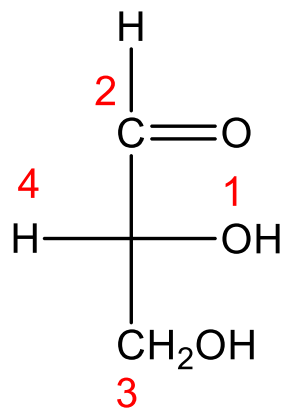
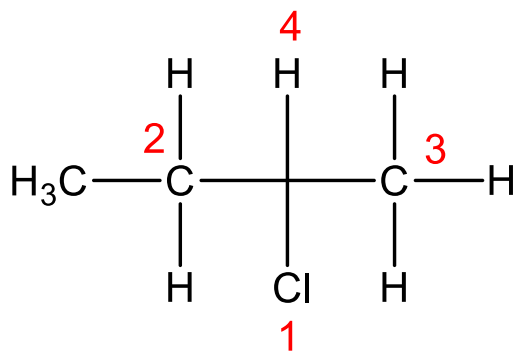


(R)-2-bromobutane



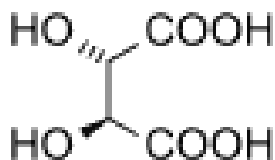
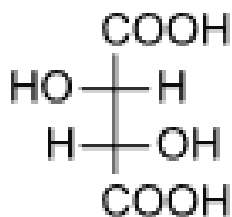
(S)- 2-hydroxy propanoic acid

Try followings

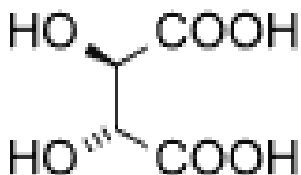
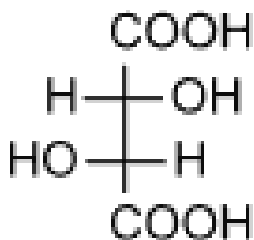


Diastereomers

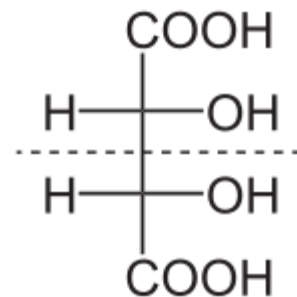
Some optical isomers may rotate the plane of polarization by different amounts, called diastereomers.



L-(+)-tartaric acid
dextrotartaric acid



D-(-)-tartaric acid
levotartaric acid

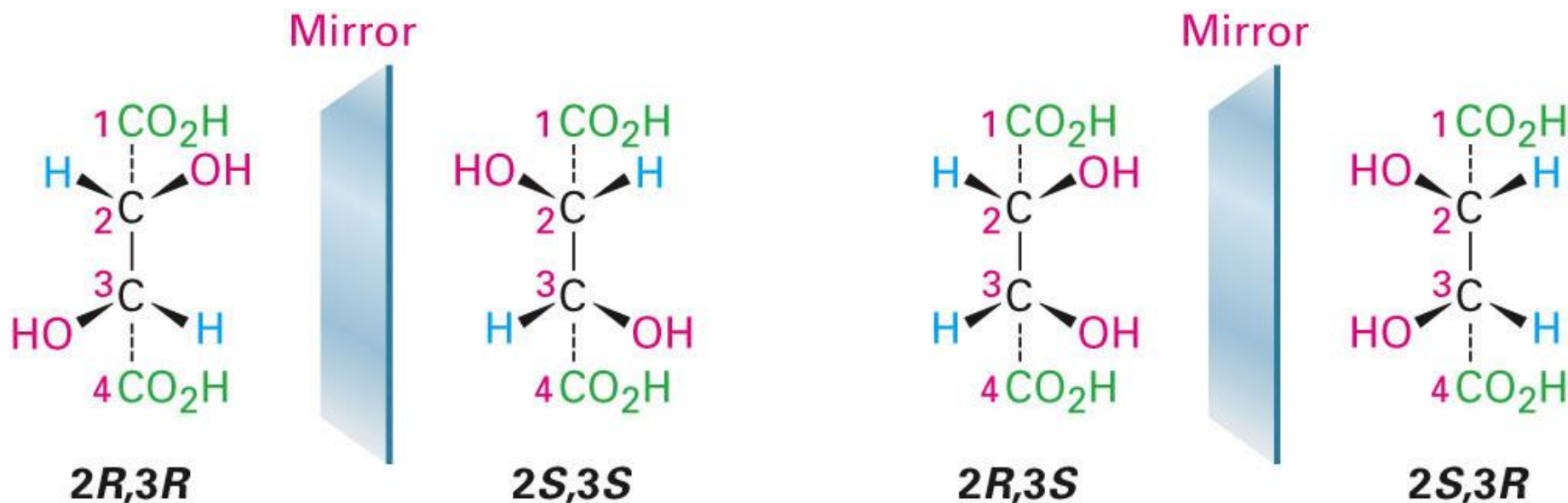


mesotartaric acid

(1:1) DL-tartaric acid "racemic acid"

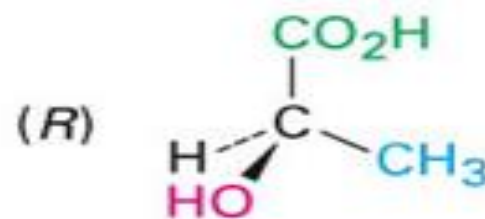
Meso Compounds

- Tartaric acid has two chiral centers and two diastereomeric forms
- One form is chiral and one is achiral, but both have two chiral centers
- An achiral compound with chiral centers is called a *meso* compound – it has a plane of symmetry
- The two structures on the right in the figure are identical so the compound (**2R, 3S**) is achiral
- Identical substitution on both chiral centers

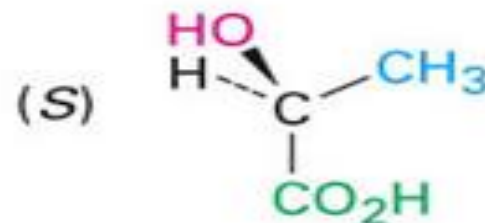


Racemic Mixtures

- A 50:50 mixture of two chiral compounds that are mirror images does not rotate light – called a **racemic mixture** (named for “racemic acid” that was the double salt of (+) and (-) tartaric acid)



+



Racemic lactic acid
(50% *R*, 50% *S*)